

GutMGS User Guidance

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Introduction:

GutMGS is a web-based calculator designed for analysis and validation of gut microbial-metabolite features and biomarkers for Myasthenia gravis (MG). GutMGS aims to simplify the microbial-metabolite data analysis through SHapley Additive exPlanations (SHAP) analysis. The model is based on our ML-SHAP model database. SDSMG offers users the capability to not only explore global explanations but also validate personalized explanations. Additionally, the platform provides data visualization tools for enhanced user readability.

About the calculator:

The main page introduces the main features of GutMGS on the left column. The left column includes the following in sequence:

1. **System Name and Authors:** Diagnosis screening for myasthenia gravis (MG) based on a machine learning model (GutMGS), authored by Che-Cheng Chang, Tzu-Chi Liu, Chi-Jie Lu, Hou-Chang Chiu, and Wei-Ning Lin.
2. **Simplified instruction for Features Input:** Please enter the data as follows: (1) Sample sequence numbers (unit: number) in the ASV table corresponding to the ASV-ID, and (2) Peak values (unit: E-08) in the MSMSID table corresponding to the HMDB-ID

The screenshot displays the GutMGS web interface. On the left, a sidebar contains the title 'Diagnosis screening for MG with gut features based on ML model (GutMGS)' and the authors' names. Below this is the 'Instructions for Feature Input' section, which provides detailed steps for data entry. On the right, the 'ML model screening result' section is visible, featuring a 'Download of user guide' button. Two red arrows with numbers 1 and 2 point to the sidebar title and the instructions section, respectively.

Diagnosis screening for MG with gut features based on ML model (GutMGS)
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Instructions for Feature Input
Please input your data as follows:
(1) Sample sequence numbers (unit: number) in the ASV table corresponding to the ASV-ID
(2) Peak values (unit: E-08) in the MSMSID table corresponding to HMDB-ID

Microbiota features (ASV table): ASV-ID

ASV_315	0
ASV_327	0
ASV_51	0
ASV_318	0
ASV_327	0

ML model screening result
We provide detailed guidance through step-by-step instructions. Users can download the file below:
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On the main page, there are 48 features in left navigation panel, including 1) 13 ASVs, 2) 35 metabolites. These 48 features are derived from our study titled 'Explainable Machine Learning Model for Identifying Key Gut Microbes and Metabolites Biomarkers Associated with Myasthenia Gravis.' Using the ML-SHAP methodology, we identified the top 1% ASV (13 SCASV) and top 15% Metabolite (35 SCM) as the most influential factors impacting MG diagnosis. Users have the option to input relevant numerical values for analysis

Prior to utilizing the system and obtaining results, users are required to utilize the experimental software (QIIME2) and reference database (HMDB database) for microbial-metabolite data transformation. This is done to standardize the input IDs and variable values. After transforming the data using the databases, users can input relevant numerical values.

3. **Important Feature Value Entry:** including sample sequence numbers (unit: number) in the ASV table corresponding to the ASV-ID (Microbiota features). Once the ASV-Number entry is completed, the corresponding total sequence number of bacteria at the taxonomic in family level will be displayed in the blue (*Lachnospiraceae*) and purple (*Ruminococcaceae*) squares below.

Microbiota features (ASV table): ASV-ID

ASV_101

0

-

+

ASV_107

0

-

+

ASV_51

0

-

+

ASV_118

0

-

+

ASV_327

0

-

+

ASV_83

0

-

+

ASV_12

0

-

+

ASV_255

0

-

+

ASV_593

0

-

+

ASV_109

0

-

+

ASV_327

0

-

+

ASV_83

0

-

+

ASV_12

0

-

+

ASV_255

0

-

+

ASV_593

0

-

+

ASV_109

0

-

+

f__Lachnospiraceae Total:0

f__Ruminococcaceae Total:0

4. **Important Feature Value Entry:** Peak values (unit: E-08) in the MSMSID table corresponding to HMDB-ID (The metabolite features)

Metabolite features (MSMSID table): HMDB-ID	
HMDB0000195 (Inosine) (Unit: E-08)	0.00
HMDB0000001 (1-Methylhistidine) (Unit: E-08)	0.00
HMDB0000715 (Kynurenic acid) (Unit: E-08)	0.00
HMDB0001065 (2-Hydroxyphenethylamine) (Unit: E-08)	0.00
HMDB0014516 (Thiethylperazine) (Unit: E-08)	0.00
HMDB0001932 (Metoprolol) (Unit: E-08)	0.00
HMDB0000673 (Linoleic acid) (Unit: E-08)	0.00
HMDB0000738 (Indole) (Unit: E-08)	0.00
HMDB0011745 (N-Acetyl-L-methionine) (Unit: E-08)	0.00
HMDB0037491 (Liquiritin aploside) (Unit: E-08)	0.00

5. **Start analysis:** To start, click the 'Analysis' button located in the bottom left corner. Initiate variable analysis and diagnosis screening. For a simple user guide, refer to the User Guide provided in the bottom-left section of the system. For a detailed user guide, click on the left-side tab to access and read comprehensive instructions."

HMDB0035473 (Avocadyne) (Unit: E-08)	0.00
HMDB0015655 (Methyltestosterone) (Unit: E-08)	0.00
HMDB0002259 (Heptadecanoic acid) (Unit: E-08)	0.00
HMDB0015185 (Novobiocin) (Unit: E-08)	0.00
Analysis	
Simple User Guide These 48 features, ID and values are derived from our study titled 'Explainable Machine Learning Model for Identifying Key Gut Microbes and Metabolites Biomarkers Associated with Myasthenia Gravis.'. Prior to utilizing the system and obtaining results, users are required to utilize the provided experimental software (QIIME2) and reference datasets (HMDB database) for data transformation. This is done to standardize the input IDs and variable values. After transforming the data using the aforementioned databases, users can input relevant numerical values. Upon obtaining the numerical outcomes, data entry facilitates the generation of SHAP results and subsequent analysis for prediction screening. The required information including (1) The ASV table, taxonomy name and sample sequence number is provided from QIIME 2. The Metabolites feature table (MSMSID table) and values is obtained from comparison with the HMDB datasets, including the HMDB-ID, compound name and peak values.	

6. **ML model screening Result:** The prediction result is displayed in the colorful chart on the right side, which includes Prediction of Diagnosis Outcomes and automatically generated SHAP Waterfall plots for personalized local explanations. Simultaneously, the prediction result provides a conclusion on Prediction of the diagnosis outcome, indicating either a Negative Diagnostic Screen for MG or a Positive Diagnostic Screen for MG. At the bottom of each SHAP Waterfall plot, the base value (0.653) is visible, representing the expected value of MG (mean model output across the training data). The sum of all SHAP values ($f(x)$) for each local explanation equals the predicted likelihood of MG diagnosis (SHAP-based personalized prediction). Red bars indicate a positive correlation, while blue bars represent a negative correlation, with the SHAP values in each plot arranged vertically in descending order (from largest to smallest).

ML model screening result

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Prediction of Diagnosis Outcomes:

Positive diagnostic screen for MG

